Alex Newbury – COMP3200 Activities & Methodology

Research & Design Methods

The bulk of this project comes from an audio synthesiser unit, which will be driven by Hall-effect sensors in a backgammon board. As this project is a largely creative one the first stage of research will focus mostly on internet blogs, forums, and similar historic projects to identify the project scope and desired features. Once the scope of the project has been found, focus will then shift to more academic sources. These will allow for the project to be broken down into manageable modules and find the required method for circuit design. Datasheets will also be invaluable in this project, as use-cases such as constructing a voltage-controlled filter from an LM13700 are standard uses for such ICs and so will be included in the manufacturer's datasheet.

Analytical Methods

Where possible simulation tools such as LTSpice can be used to design and test circuits before involving any physical components, to both ensure there is no risk of damage to the components and keep project costs down as minimal spares will be needed. However, some components such as the CEM3340 Voltage Controlled Oscillator do not have publicly available models and so specific functionality will have to be explored physically, using signal generators and an oscilloscope.

After simulations have been completed and correct functionality verified, circuits will then be built onto breadboard. An oscilloscope and signal generators can be used to fully test circuit behaviours before integrating any parts of the system, as this will save time later in the project by allowing for the identification and resolution of any issues with the designs before either committing the design to stripboard. The final version of the system will be built onto stripboard - as this is a creative piece the numbers required to make the manufacture of PCBs cost-effective would likely not be sold.

Environmental Concerns and Health & Safety

The heath and safety risks of the project have been assessed against standard lab risk assessments. The standard precautions should be taken when using a soldering iron or power supplies, but apart from these regular pieces of lab equipment there are no concerning safety aspects of the project. A benchtop power supply will be used to power the system with +/-15V, and the terminals for these connections should be clearly labelled to avoid incorrect connections.

One of the stretch goals within the project is to package the design into a table (see PERT diagram or list of milestones). There are environmental considerations to be given to the sourcing of wood used for the project - FSC certified wood is confirmed to be environmentally sustainably sourced, or second-hand/reclaimed timber could be used as a lower-carbon alternative.